

7. Further discussions

1) The $\vec{B}$ field of a long straight wire

Find the magnetic field $\vec{B}$ generated by a long straight wire, carrying a current I .

Solution 1:

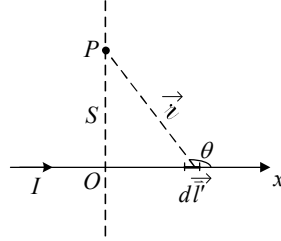


Figure. Magnetic field generated by a long straight wire: definition of parameters.

Directly applying the Biot-Savart Law,

$$\vec{B} = \frac{\mu_0}{4\pi} I \int \frac{d\vec{l}' \times \hat{r}}{r^2}$$

Denote the distance between the field point P and the straight wire to be s , $d\vec{l}' = \hat{x}dx$, $\hat{x} \times \hat{r} = \sin\theta \hat{y}$ ($\hat{y}$ points out of the screen), we can compute the magnitude of B by:

$$\begin{aligned} B(s) &= \frac{\mu_0}{4\pi} I \int_{-\infty}^{\infty} \frac{\sin\theta}{(x^2 + s^2)} dx = \frac{\mu_0}{4\pi} I \int_{-\infty}^{\infty} \frac{s}{(x^2 + s^2)^{3/2}} dx \\ &= \frac{\mu_0 I s}{4\pi} \int_{-\infty}^{\infty} \frac{dx}{(x^2 + s^2)^{3/2}} = \frac{\mu_0}{4\pi} I s \left. \frac{x}{s^2 \sqrt{s^2 + x^2}} \right|_{-\infty}^{+\infty} \end{aligned}$$

Finally, we get the expression for the magnetic field:

$$\vec{B}(s) = \frac{\mu_0 I}{2\pi s} \hat{\phi}$$

Solution 2:

The symmetry of the problem (in the azimuthal direction) allows us to apply Ampere's Law:

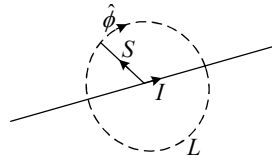


Figure. Apply Ampere's Law to solve the problem.

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I \Rightarrow 2\pi S \cdot B = \mu_0 I$$

$$\vec{B} = \frac{\mu_0 I}{2\pi S} \hat{\phi}$$

2) The $\vec{B}$ field of a long solenoid

A solenoid is a tightly wound coil of wire with n turns per unit length around a cylindrical core. The wire carries a current I , and due to the tight winding, a surface current density $\vec{K} = nI\hat{\phi}$ flows along the cylinder's surface. A section of the solenoid is illustrated in the figure below. The influence of any longitudinal current component (along the z -axis) is neglected. Find $\vec{B}$ and $\vec{A}$ distributions.

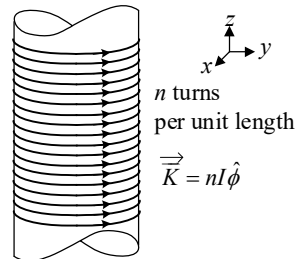


Figure. A section of the tightly wound solenoid.

Solution:

The generated magnetic field has no $\hat{\phi}$ component. This can be demonstrated by performing a loop integral around a path that is parallel and concentric to $\vec{K}$. Since this loop encloses no net current, Ampere's Law dictates that the integral must be zero, confirming the absence of a $\hat{\phi}$ component in the magnetic field.

The magnetic field has no $\hat{\rho}$ component (radial direction) either. Suppose, for contradiction, that the solenoid generates a radial field pointing outward. According to the Biot-Savart Law, reversing the current direction would then produce a radial field pointing inward. However, reversing the current is physically equivalent to flipping the solenoid upside-down, which should not alter the direction of the radial component. This contradiction can only be resolved if the radial component is, in fact, zero.

Thus, the only surviving component of the magnetic field is along the axis of the solenoid ($\hat{z}$ component). To determine its value outside the solenoid, we integrate the magnetic field along a rectangular path in the radial plane (at a fixed azimuthal angle). The rectangle has a height of l , as illustrated in the following figure. The axial component of the magnetic field should be uniform due to uniformity of the solenoid along $\hat{z}$.

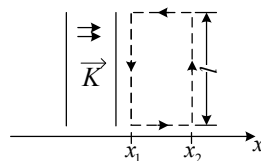


Figure. Side-view of the Amperian loop outside the solenoid.

The loop encloses no current, and the two edges along the radial direction do not contribute to the integral. Consequently, the integration result is:

$$\oint_L \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc} = (B(x_1) - B(x_2))\ell = 0.$$

At this point, we have reached a key milestone: the magnetic field outside the solenoid is zero, and all the magnetic flux is entirely confined within the solenoid, aligned along the $\hat{z}$ direction.

To determine the magnetic field inside the solenoid, we evaluate a similar line integral around the Amperian loop in the radial plane, this time extending across the wall of the solenoid's wall, as illustrated below.

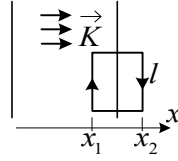


Figure. Side-view of the Amperian loop traversing the solenoid's wall.

The loop encloses a current $I_{enc} = I n \ell$, and the two edges along the radial direction do not contribute to the integral:

$$\oint_L \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc} = \mu_0 I n \ell = B(x_1)\ell.$$

Thus, it is straightforward to find the magnetic field inside the solenoid:

$$\boxed{\vec{B} = \mu_0 I n \hat{z}}.$$

This is an amazing result, as the surface current around the solenoid works collectively to establish a uniform magnetic field inside while perfectly canceling the field outside.

Now, we proceed to compute the vector potential distribution. Recall that we have established that the relationship between $\vec{A}$ and $\vec{B}$ mirrors that between $\vec{B}$ and $\vec{J}$. In this case, since the distribution of $\vec{B}$ is along a straight wire, the distribution of $\vec{A}$ should be in the $\hat{\phi}$ direction (refer to example 1). The symmetry in the azimuthal direction allows us to determine A through a line integral along the Amperian loop, as shown in the following figure.

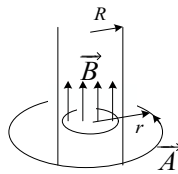


Figure. Amperian loops used for calculating the vector potential.

Performing the line integral, we get:

$$\oint \vec{A} \cdot d\vec{\ell} = \Phi_B,$$

where Φ_B is the total enclosed magnetic flux.

For the loop outside the solenoid (i.e., when the radius of the loop r exceeds that of the solenoid R), the integral leads to $2\pi r A = \mu_0 I n \pi R^2$, thus the vector potential outside the solenoid is:

$$\vec{A}_{out} = \frac{1}{2} \frac{\mu_0 I n R^2}{r} \hat{\phi}.$$

If the loop is inside the solenoid (i.e., $r < R$), the integral yields $2\pi r A = \mu_0 I n \pi r^2$, and we obtain the vector potential inside the solenoid:

$$\vec{A}_{in} = \frac{1}{2} \mu_0 I n r \hat{\phi}.$$

The vector potential's distribution along the radial direction is illustrated in the following figure.

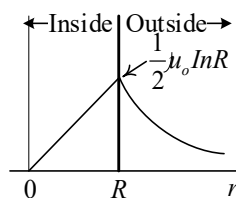


Figure. Distribution of A as a function of r .

3) After inserting a material with magnetic susceptibility of χ_m into the solenoid, compute the $\vec{B}$ field.

As in the previous example, we can perform a loop integration across the boundary of the solenoid, however we have no idea about the total current distribution. In this case, we apply Ampere's Law for H , which only involves free current: $\oint \vec{H} \cdot d\vec{\ell} = I_{free,enc}$, from which we readily obtain $H = n I$. Knowing the magnetic susceptibility, we can compute $\vec{B}$ based on $\vec{H}$:

$$\vec{B} = \mu_0 (1 + \chi_m) \vec{H} = \mu_0 (1 + \chi_m) n I \hat{z}.$$

4) Compute the magnetic vector potential due to a spinning spherical shell which has a uniform surface charge density distribution σ .

We start the derivation from the fundamental definition of $\vec{A}$:

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}(\vec{r}')}{z} da'$$

To aid our analysis, the problem is illustrated below.

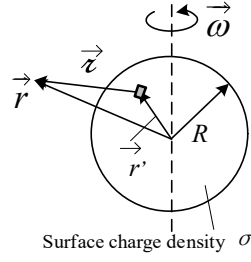


Figure. Illustration of the spinning charge-carrying sphere.

The movement of the surface charge generates a surface current, which can be computed by

$$\vec{K}(\vec{r}') = \sigma(\vec{r}')\vec{v}(\vec{r}') = \sigma\vec{\omega} \times \vec{r}'$$

Substituting the above expression into the formula of $\vec{A}$:

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\sigma\vec{\omega} \times \vec{r}'}{r} d\vec{a}' = \frac{\mu_0\sigma\vec{\omega}}{4\pi} \times \int \frac{\vec{r}'}{r} d\vec{a}'$$

Set $\vec{F} = \int \frac{\vec{r}'}{r} d\vec{a}'$ which is a vector along $\vec{r}$ (the integration is symmetric with respect to $\vec{r}$):

$$\vec{F} = F\hat{r}.$$

According to the generating function of the Legendre polynomials (which we encountered during the derivation of the magnetic dipole's vector potential):

(a) Inside the sphere: $|\vec{r}| < R$, $\frac{1}{r} = \sum_{n=0}^{\infty} \frac{r^n}{R^{n+1}} P_n(\cos \theta)$, the quantity F can be evaluated as:

$$F = \int_0^{2\pi} \int_0^{\pi} R \cos \theta \sum_{n=0}^{\infty} \frac{r^n}{R^{n+1}} P_n(\cos \theta) R^2 \sin \theta d\theta d\varphi,$$

Employing $P_1(\cos \theta) = \cos \theta$ and the orthogonality condition

$$\int_0^{\pi} P_n(\cos \theta) P_m(\cos \theta) \sin \theta d\theta = \int_{-1}^1 P_n(x) P_m(x) dx = \frac{2}{2n+1} \delta_{nm}$$

where δ_{nm} is the Kronecker delta, which is 1 if $n = m$ and 0 otherwise, we conclude that

$$F = R \int_0^{2\pi} r \cdot \frac{2}{3} d\varphi = \frac{4}{3} \pi R \cdot r,$$

And consequently,

$$\vec{A} = \frac{\mu_0\sigma}{4\pi} \cdot \frac{4}{3} \pi R \vec{\omega} \times \vec{r} = \frac{\mu_0\sigma R}{3} \vec{\omega} \times \vec{r}.$$

(b) Outside the sphere: $|\vec{r}| > R$, $\frac{1}{r} = \sum_{n=0}^{\infty} \frac{R^n}{r^{n+1}} P_n(\cos \theta)$

$$F = \int_0^{2\pi} \int_0^\pi R \cdot P_1(\cos \theta) \sum_{n=0}^{\infty} \frac{R^n}{r^{n+1}} P_n(\cos \theta) R^2 \sin \theta \, d\theta d\varphi$$

With the same mathematical treatment, we conclude that

$$\vec{A} = \frac{\mu_0 \sigma R^4}{3r^3} \vec{\omega} \times \vec{r}.$$

Comparison with the magnetic dipole potential reveals that the spinning sphere produces a magnetic field identical to that of a dipole.

5) The Aharonov-Bohm effect (why $\vec{A}$ is more fundamental than $\vec{B}$)

In Newtonian dynamics and classical electromagnetism, forces are considered fundamental, and the electric and magnetic fields E and B are defined accordingly. The scalar and vector potentials are introduced primarily as mathematical conveniences to simplify the calculation of these fields. However, in quantum mechanics, this perspective shifts: the fields E and B gradually lose their primacy, while the potentials emerge as physically significant quantities. This is most strikingly demonstrated by the Aharonov-Bohm effect shown below.

In quantum mechanics, matters—such as electrons—exhibits wave-like behavior, much like photons. When an electron interacts with a magnetic field, this interaction manifests as an additional phase shift in its wavefunction, given by:

$$\varphi = \frac{e}{\hbar} \int_{path} \vec{A} \cdot d\vec{\ell}.$$

In this expression, e is the electron charge, $\hbar$ is the reduced Planck constant, $\vec{A}$ is the magnetic vector potential, and the integral is evaluated along the path taken by the electron as it moves through the magnetic field.

Astonishingly, due to the wave-like nature of electrons, it is possible to reproduce Young's double-slit experiment using electrons, as illustrated in the figure (left). When electrons are emitted one by one (unlike photons, electrons experience mutual repulsion due to the Coulomb force—so it's best not to send them all at once!), they gradually build up an interference pattern on the detection screen.

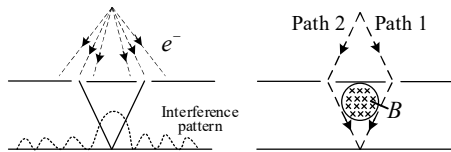


Figure. The electron double-slit experiment (left) and the Aharonov-Bohm-modified version (right).

The positions of the interference peaks and valleys are determined by the phase difference between the two interfering paths. For example, if the paths are perfectly symmetric, the central interference peak appears at the center of the setup. In the experiment proposed by Aharonov and Bohm, a modified version of the classic double-slit experiment was designed. A solenoid was placed between the two electron paths, as illustrated in the figure (right). As discussed in Example 3, the magnetic field is confined entirely within the solenoid and is not directly "seen" by the electrons. However, the magnetic vector potential $\vec{A}$ outside the solenoid is not zero along the electron paths. If the vector potential indeed imparts a phase shift to the electrons, the two paths become asymmetric, leading to a shift in the interference pattern. The resulting phase difference, as discussed above, is given by:

$$\Delta\varphi = \frac{e}{\hbar} \int_{\text{Path 1}} \vec{A} \cdot d\vec{\ell} - \frac{e}{\hbar} \int_{\text{Path 2}} \vec{A} \cdot d\vec{\ell}.$$

The minus sign of the second term on the right-hand side is equivalent to reversing the integration direction along path 2, so that path 1 and (the reversed) path 2 together form a closed loop:

$$\Delta\varphi = \frac{e}{\hbar} \oint_L \vec{A} \cdot d\vec{\ell},$$

where the loop along the dashed line shown in the figure encloses all the magnetic flux Φ_B :

$$\Delta\varphi = \frac{e}{\hbar} \Phi_B.$$

As soon as a current is applied to the solenoid—turning on the magnetic field—a phase shift is observed, precisely matching the expression given above. It is important to note that the electrons travel through a region where the magnetic field is zero. Therefore, the shift in the interference pattern is not caused by the Lorentz force, but arises more fundamentally from the presence of the vector potential. This phenomenon is purely quantum mechanical and cannot be explained by classical electromagnetic theory, which attributes interactions with charged particles to the fields rather than to the potentials.

6) Electromagnetic forces on electric and magnetic dipoles

① Electric force on an electric dipole

An electric dipole placed in a non-uniform electrostatic field will experience a force. To analyze this behavior, we begin by considering the dipole's electrostatic energy. A dipole can be constructed by bringing two oppositely charged particles from infinity and placing them at a fixed separation. The negative charge, $-q$, is placed at $\vec{r}$, while the positive charge, $+q$, is placed at $\vec{r} + \vec{d}$, while $\vec{d}$ is the

vector representing the separation between the charges. The figure below illustrates the definitions of the relevant vectors and the final configuration of the assembled dipole.

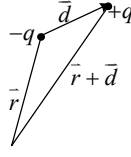


Figure. Vectors that define the final position of the dipole.

The energy involved in this process can be found by:

$$\begin{aligned}
 U_e &= q\phi(\vec{r} + \vec{d}) - q\phi(\vec{r}) = q \left[- \int_{\vec{r}}^{\vec{r} + \vec{d}} \vec{E}(\vec{r}') \cdot d\vec{l}' \right] \\
 &= -q\vec{E} \cdot \vec{d} \quad (\vec{E} \text{ is almost constant between } \vec{r} \text{ and } \vec{r} + \vec{d}) \\
 &= -q \underbrace{\vec{d}}_{\vec{p}} \cdot \vec{E} \\
 &= -\vec{p} \cdot \vec{E}
 \end{aligned}$$

The electric force on the dipole is the negative gradient of the potential energy:

$$\vec{F}_e = -\nabla U_e = \nabla(\vec{p} \cdot \vec{E})$$

② Magnetic force on a magnetic dipole

The force on a magnetic dipole is significantly more challenging to derive than that on an electric dipole. To simplify the derivation while retaining the physical insight, we consider a magnetic dipole formed by a square current loop, as illustrated in the following figure. The dipole loop lies in the yz -plane, with its dipole moment pointing along $\hat{x}$.

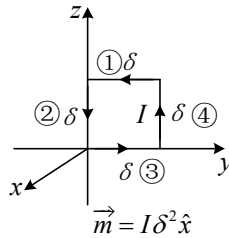


Figure. The dipole formed by a square current loop.

First, we compute the total force on edge ① and ③:

$$\vec{F}_{1,3} = I\delta(-\hat{y}) \times \vec{B}_1 + I\delta\hat{y} \times \vec{B}_3$$

$$\begin{aligned}
&= I\delta(-\hat{y}) \times (\vec{B}_1 - \vec{B}_3) \\
&\approx -I\delta\hat{y} \times \frac{\partial \vec{B}}{\partial z}\delta \\
&= -I\delta^2\hat{y} \times \partial_z \vec{B} = -|\vec{m}|\hat{y} \times \partial_z \vec{B}
\end{aligned}$$

In the above equation, $\vec{m} = I\delta^2\hat{x}$ is the magnetic dipole moment. Similarly, the total force on edge ② and ④ can be computed as $\vec{F}_{2,4} = |\vec{m}|\hat{z} \times \partial_y \vec{B}$. Combining these results, the total force on the dipole is:

$$\begin{aligned}
\vec{F}_m &= m \left\{ \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ \partial_y B_x & \partial_y B_y & \partial_y B_z \end{vmatrix} - \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 1 & 0 \\ \partial_z B_x & \partial_z B_y & \partial_z B_z \end{vmatrix} \right\} \\
&= m \{ \hat{y} \partial_y B_x \quad \underbrace{-\hat{x} \partial_y B_y - \hat{x} \partial_z B_z}_{\hat{x} \partial_x B_x,} \quad + \hat{z} \partial_z B_x \} \\
&\quad \text{since } \partial_x B_x + \partial_y B_y + \partial_z B_z = 0 \text{ (}\nabla \cdot \vec{B} = 0\text{)} \\
&= m \{ \hat{x} \partial_x B_x + \hat{y} \partial_y B_x + \hat{z} \partial_z B_x \} \\
&= m \nabla B_x = \nabla(m B_x) \\
&\xrightarrow[\substack{\vec{m} = m\hat{x}, \\ \vec{B} = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}}]{\hspace{1cm}} \nabla(\vec{m} \cdot \vec{B})
\end{aligned}$$

Thus, we conclude that the magnetic force on a magnetic dipole is given by $\nabla(\vec{m} \cdot \vec{B})$. It is important to note that the above analysis is not a rigorous proof, but rather an illustrative explanation based on a simple example.